Self-attention: each headline takes information from the others

Follow headline A through one news bucket. The same operation is repeated for every headline, with its own set of weights.

1 / Start with numerical headline representations

Text and sentiment have already been fused into **16 numbers per headline** in our model. The headlines below are invented for this explanation.

A / Apple earnings improve
Encoded as 16 numbers

B / iPhone sales increase
Encoded as 16 numbers

C / Oil prices decline
Encoded as 16 numbers

2 / Compare A with every headline; turn scores into weights

Learned transformations make a **query** for A and a **key** for each headline. Their numerical comparisons produce scores.

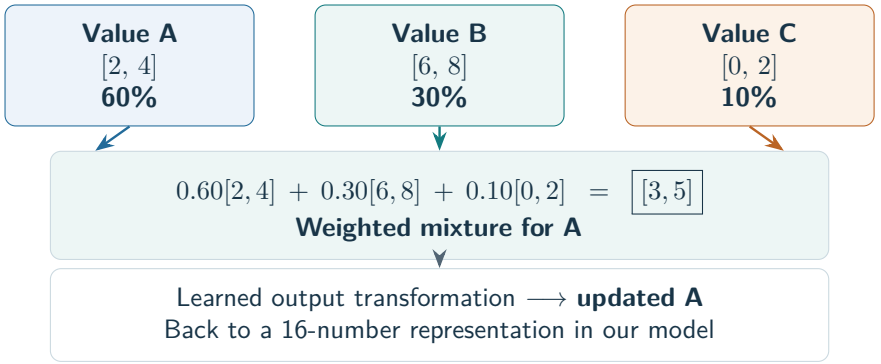
Comparison	Example score	Weight
A's query with A's key	1.79	60%
A's query with B's key	1.10	30%
A's query with C's key	0.00	10%

Softmax converts scores into positive weights that add up to **100%**. Higher scores receive more weight. Numbers above are rounded, illustrative outputs.

The comparison uses learned query/key vectors, not a hand-written rule saying which financial stories should be related.

3 / Use the weights to mix information

A separate learned transformation makes each headline's **value vector**: the numerical information to mix. Use two numbers here just to make the arithmetic visible.



Say it simply: "Each headline compares itself with the group. Those scores determine how much information it takes from each headline, including itself. The weighted mixture becomes its updated representation."

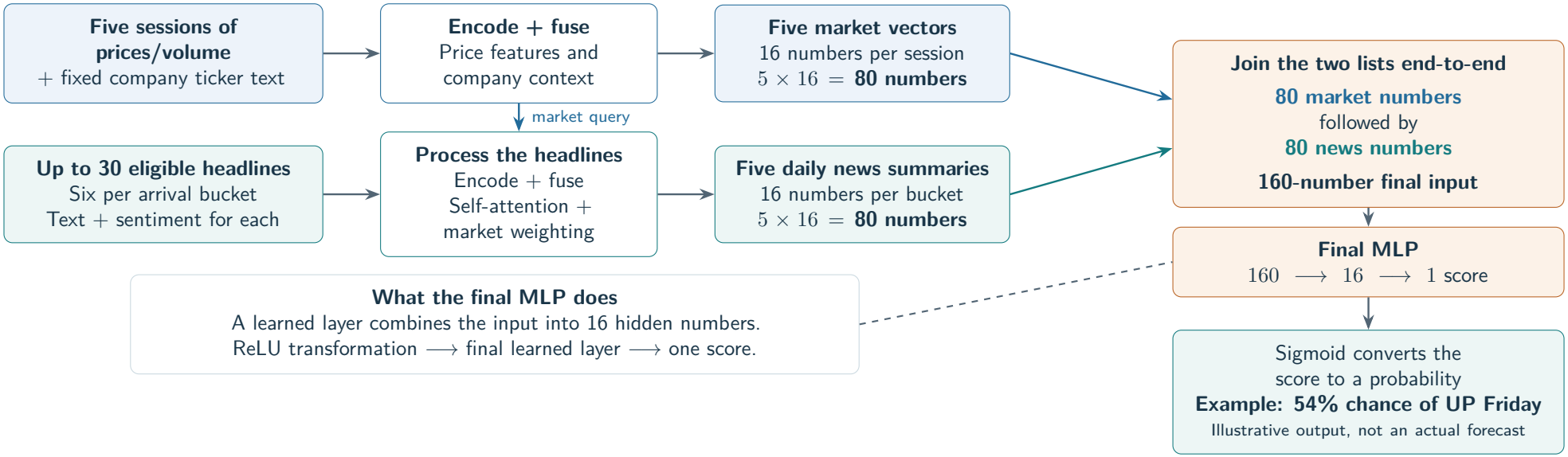
Where do the rules come from? Training errors adjust the query, key, value and output transformations. Attention weights are then recalculated for each input group.

What happens next? A *second* attention step uses market context to weight these updated vectors into a daily news summary. These are different weights from the ones above.

Within-bucket, single-head illustration of our plain self-attention path. These weights are illustrative, not measured relevance or causal importance. Interactions do not guarantee better forecasts.

The final MLP receives numerical summaries of market and news

Thursday example: after Thursday's close, use the previous Friday, Monday, Tuesday, Wednesday and Thursday to predict whether **Friday closes higher than Thursday**. Assume no holidays.



FININ and our two versions: same idea, different news arrangement

Version	Market block received	News block received	Final input size
FININ paper, Eq. 7	Encoded market vector for each input day	Weighted news-summary vector for each input day	Depends on lookback and width
Our baseline (diagram above)	Five vectors: $5 \times 16 = 80$	Five summaries: $5 \times 16 = 80$	**160 numbers**
Our recent pooled/decay tests	Same five vectors: **80**	Pool the five buckets into **one 16-number summary**	**96 numbers**

Say it simply: "The MLP receives a long list of market numbers and summarized news numbers. Headline text, sentiment and attention weights have already shaped the news summary; they are not separate raw inputs to this final step."

Timing: news must be eligible by Thursday's cutoff. Our date-based rule assumes availability from source date $D + 2$; Tuesday-dated news can therefore enter Thursday's bucket. Exact historical availability remains uncertain.

FININ market context describes the index; ours uses fixed company ticker context. Actual widths verified against local model code. Reference: Wang, Cohen & Ma (2024), Sections 4.1–4.3, Eq. 7.